

A Preferred-Habitat Captive-Demand Model: Revised Statement with Approximation, Consumption, and Mandate Corrections

Lucas Leturia (revised technical statement)

May 2026

Abstract

This note restates the preferred-habitat captive-demand model in a form that keeps the clean algebra of the original derivation while making explicit the approximation order, the common-consumption convention used for closed-form portfolio shares, the consumption-ratio correction to those shares, and the sufficient conditions under which a market-value mandate raises the captive agent's physical long-bond quantity and lowers the unconstrained agent's long-bond portfolio share. The global monotonicity result is deliberately held out as requiring additional restrictions on price elasticity and wealth responses.

1 Environment

Time is discrete, $t = 0, 1, 2, \dots$. There is a single non-storable consumption good. Aggregate resources satisfy

$$\mu_A c_t^A + \mu_B c_t^B + g_t = y_t, \quad (1)$$

where g_t is exogenous government spending, y_t is an exogenous endowment, and $\mu_A + \mu_B = 1$.

The nominal price level is P_t , and gross inflation is

$$\pi_t \equiv \frac{P_t}{P_{t-1}}. \quad (2)$$

Lowercase quantities are real unless explicitly stated otherwise.

The government issues two nominal bonds:

- a one-period nominal short bond paying one unit of currency at $t + 1$, with nominal stock B_t^S , real stock $b_t^S \equiv B_t^S / P_t$, and real price Q_t^S ;
- a geometric perpetuity, or long bond, with decay $\delta \in [0, 1)$. A unit issued at t pays nominal coupons $1, \delta, \delta^2, \dots$ at dates $t + 1, t + 2, t + 3, \dots$. Its nominal stock in fresh-bond-equivalent units is B_t^L , its real stock is $b_t^L \equiv B_t^L / P_t$, and its real price is Q_t^L .

The maturity-composition rule is exogenous:

$$b_t^L = \chi_t (b_t^S + b_t^L), \quad \chi_t \in [0, 1]. \quad (3)$$

Agent $i \in \{A, B\}$ holds $b_t^{S,i}$ and $b_t^{L,i}$ in real per-capita terms. Market clearing requires

$$\mu_A b_t^{S,A} + \mu_B b_t^{S,B} = b_t^S, \quad \mu_A b_t^{L,A} + \mu_B b_t^{L,B} = b_t^L. \quad (4)$$

2 Domain and admissibility restrictions

The following restrictions are imposed throughout the local derivations.

Assumption 1 (Domain). *For each $i \in \{A, B\}$ and all relevant dates,*

$$c_t^i > 0, \quad Q_t^S > 0, \quad Q_t^L > 0. \quad (5)$$

Individual portfolio wealth is strictly positive:

$$W_t^i \equiv b_t^{S,i} + b_t^{L,i} > 0, \quad (6)$$

so that the long-bond portfolio share

$$\alpha_t^i \equiv \frac{b_t^{L,i}}{W_t^i} \quad (7)$$

is well-defined. Local sign and comparative-static results are stated around interior allocations with

$$\alpha_t^i \in (0, 1). \quad (8)$$

Assumption 2 (Admissible portfolio paths). *For transversality arguments, admissible equilibrium paths satisfy either natural debt limits or bounded gross real bond positions relative to the endowment. A convenient sufficient condition is that there exists finite C such that*

$$|Q_t^S b_t^{S,i}| + |Q_t^L b_t^{L,i}| \leq C y_t \quad (9)$$

for each i and all t along admissible paths.

Remark 1. *The bounded-gross-position condition is not implied by the household budget constraint alone. It is an admissibility restriction needed to rule out offsetting large long and short positions, especially for the constrained agent in the binding mandate regime.*

3 Preferences and habitat costs

Each agent $i \in \{A, B\}$ maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(c_t^i)^{1-\sigma}}{1-\sigma} - H_i(b_t^{S,i}, b_t^{L,i}) \right], \quad (10)$$

where $\beta \in (0, 1)$ and $\sigma > 0$ are common across agents. The habitat penalty is

$$H_i(b_t^{S,i}, b_t^{L,i}) \equiv \frac{\phi}{2} (\alpha_t^i - \theta_i)^2 W_t^i, \quad (11)$$

where $\phi > 0$, $\theta_i \in [0, 1]$, and

$$W_t^i = b_t^{S,i} + b_t^{L,i}, \quad \alpha_t^i = \frac{b_t^{L,i}}{W_t^i}. \quad (12)$$

Heterogeneity is encoded by

$$\theta_A < \theta_B, \quad (13)$$

so that A is the short-habitat type and B is the long-habitat type.

4 Household budget constraint and first-order conditions

Agent i faces the real flow budget constraint

$$c_t^i + Q_t^S b_t^{S,i} + Q_t^L b_t^{L,i} = y_t^i - \tau_t + \frac{b_{t-1}^{S,i}}{\pi_t} + \frac{(1 + \delta Q_t^L) b_{t-1}^{L,i}}{\pi_t}. \quad (14)$$

Let λ_t^i denote the multiplier on the budget constraint. The consumption FOC is

$$\lambda_t^i = (c_t^i)^{-\sigma}. \quad (15)$$

Define the one-period stochastic discount factor

$$M_{t+1}^i \equiv \beta \left(\frac{c_{t+1}^i}{c_t^i} \right)^{-\sigma}. \quad (16)$$

The useful intermediate derivatives are

$$\frac{\partial W_t^i}{\partial b_t^{S,i}} = 1, \quad \frac{\partial W_t^i}{\partial b_t^{L,i}} = 1, \quad (17)$$

$$\frac{\partial \alpha_t^i}{\partial b_t^{S,i}} = -\frac{\alpha_t^i}{W_t^i}, \quad \frac{\partial \alpha_t^i}{\partial b_t^{L,i}} = \frac{1 - \alpha_t^i}{W_t^i}. \quad (18)$$

Let

$$\varepsilon_t^i \equiv \alpha_t^i - \theta_i. \quad (19)$$

The exact habitat derivatives are

$$\frac{\partial H_i}{\partial b_t^{S,i}} = \frac{\phi}{2} (\varepsilon_t^i)^2 - \phi \varepsilon_t^i \alpha_t^i, \quad (20)$$

$$\frac{\partial H_i}{\partial b_t^{L,i}} = \frac{\phi}{2} (\varepsilon_t^i)^2 + \phi \varepsilon_t^i (1 - \alpha_t^i). \quad (21)$$

5 Approximation convention and habitat wedges

The clean algebra uses leading-order wedge objects and keeps the exact second-order remainder explicit.

Define

$$\xi_t^{S,i} \equiv \frac{\phi \varepsilon_t^i \alpha_t^i}{(c_t^i)^{-\sigma}}, \quad (22)$$

$$\xi_t^{L,i} \equiv -\frac{\phi \varepsilon_t^i (1 - \alpha_t^i)}{(c_t^i)^{-\sigma}} = \frac{\phi (\theta_i - \alpha_t^i) (1 - \alpha_t^i)}{(c_t^i)^{-\sigma}}. \quad (23)$$

Then

$$-\frac{1}{(c_t^i)^{-\sigma}} \frac{\partial H_i}{\partial b_t^{S,i}} = \xi_t^{S,i} + R_t^{S,i}, \quad (24)$$

$$-\frac{1}{(c_t^i)^{-\sigma}} \frac{\partial H_i}{\partial b_t^{L,i}} = \xi_t^{L,i} + R_t^{L,i}, \quad (25)$$

where the exact residuals are

$$R_t^{S,i} = R_t^{L,i} = -\frac{\phi}{2} \frac{(\varepsilon_t^i)^2}{(c_t^i)^{-\sigma}} = O((\varepsilon_t^i)^2). \quad (26)$$

Equivalently, in strict first-order form one may replace α_t^i by θ_i inside the wedge coefficients:

$$\xi_t^{S,i} = \frac{\phi \theta_i \varepsilon_t^i}{(c_t^i)^{-\sigma}} + O((\varepsilon_t^i)^2), \quad (27)$$

$$\xi_t^{L,i} = -\frac{\phi(1-\theta_i)\varepsilon_t^i}{(c_t^i)^{-\sigma}} + O((\varepsilon_t^i)^2). \quad (28)$$

The derivations below use the α_t^i -coefficient wedge convention for compact algebra, especially in the determinant representation of the fiscal rent. Final identities include the residual term where appropriate.

The Euler equations are therefore

$$Q_t^S = \mathbb{E}_t \left[\frac{M_{t+1}^i}{\pi_{t+1}} \right] + \xi_t^{S,i} + R_t^{S,i}, \quad (29)$$

$$Q_t^L = \mathbb{E}_t \left[\frac{M_{t+1}^i(1 + \delta Q_{t+1}^L)}{\pi_{t+1}} \right] + \xi_t^{L,i} + R_t^{L,i}. \quad (30)$$

At leading order, the residuals are dropped and the equations reduce to the clean wedge representation.

6 Unconstrained equilibrium

In the unconstrained regime both agents hold both bonds at interior positions. Common prices imply that the Euler equations hold for both $i = A, B$.

At leading order, differencing the long-bond Euler equation across agents gives

$$0 = \mathbb{E}_t \left[\frac{(M_{t+1}^A - M_{t+1}^B)(1 + \delta Q_{t+1}^L)}{\pi_{t+1}} \right] + \xi_t^{L,A} - \xi_t^{L,B}. \quad (31)$$

In a deterministic steady state, $M^A = M^B = \beta$, so the expectation term vanishes and

$$\xi^{L,A} = \xi^{L,B}. \quad (32)$$

Using the leading-order long-bond wedge,

$$(\theta_A - \alpha^A)(1 - \alpha^A)(c^A)^\sigma = (\theta_B - \alpha^B)(1 - \alpha^B)(c^B)^\sigma. \quad (33)$$

Define wealth shares

$$\omega_i \equiv \frac{\mu_i W^i}{\mu_A W^A + \mu_B W^B}, \quad \omega_A + \omega_B = 1, \quad (34)$$

and the wealth-weighted average habitat

$$\bar{\theta} \equiv \omega_A \theta_A + \omega_B \theta_B. \quad (35)$$

Aggregation implies

$$\omega_A \alpha^A + \omega_B \alpha^B = \chi. \quad (36)$$

Writing

$$\alpha^i = \theta_i + \varepsilon^i, \quad (37)$$

we have

$$\omega_A \varepsilon^A + \omega_B \varepsilon^B = \chi - \bar{\theta}. \quad (38)$$

6.1 Clean common-consumption algebra

For algebraic transparency, impose the local common-consumption convention

$$c^A = c^B. \quad (39)$$

Then the first-order wedge-equality condition becomes

$$\varepsilon^A(1 - \theta_A) = \varepsilon^B(1 - \theta_B). \quad (40)$$

Solving jointly with aggregation gives

$$\varepsilon^A = \frac{(\chi - \bar{\theta})(1 - \theta_B)}{\omega_A(1 - \theta_B) + \omega_B(1 - \theta_A)}, \quad (41)$$

$$\varepsilon^B = \frac{(\chi - \bar{\theta})(1 - \theta_A)}{\omega_A(1 - \theta_B) + \omega_B(1 - \theta_A)}. \quad (42)$$

6.2 Consumption-ratio correction

Without common consumption, the first-order wedge-equality condition is

$$\varepsilon^A(1 - \theta_A)(c^A)^\sigma = \varepsilon^B(1 - \theta_B)(c^B)^\sigma. \quad (43)$$

The corrected solution is

$$\varepsilon^A = \frac{(\chi - \bar{\theta})(1 - \theta_B)(c^B)^\sigma}{\omega_A(1 - \theta_B)(c^B)^\sigma + \omega_B(1 - \theta_A)(c^A)^\sigma}, \quad (44)$$

$$\varepsilon^B = \frac{(\chi - \bar{\theta})(1 - \theta_A)(c^A)^\sigma}{\omega_A(1 - \theta_B)(c^B)^\sigma + \omega_B(1 - \theta_A)(c^A)^\sigma}. \quad (45)$$

These formulas reduce to the common-consumption expressions when $c^A = c^B$.

6.3 Unconstrained long-bond price

In deterministic steady state, using agent A 's leading-order long-bond Euler equation,

$$Q^{L,\text{unc}} = \frac{\beta/\pi + \xi^{L,A,\text{unc}}}{1 - \beta\delta/\pi}. \quad (46)$$

When $\pi = 1$,

$$Q^{L,\text{unc}} = \frac{\beta}{1 - \beta\delta} + \frac{\xi^{L,A,\text{unc}}}{1 - \beta\delta}. \quad (47)$$

7 Regulatory mandate

The regulator imposes a market-value mandate on agent B 's long-bond portfolio:

$$Q_t^L \mu_B b_t^{L,B} \geq \lambda_t^L V_t^{\text{liab}}. \quad (48)$$

Let

$$m_t \equiv \lambda_t^L V_t^{\text{liab}}. \quad (49)$$

Assumption 3 (Bounded mandate). *There exists finite $\bar{M}$ such that*

$$\frac{m_t}{y_t} \leq \bar{M} \quad (50)$$

almost surely for all t .

When the mandate binds,

$$Q_t^L \mu_B b_t^{L,B} = m_t, \quad (51)$$

and therefore

$$\mu_B b_t^{L,B} = \frac{m_t}{Q_t^L}. \quad (52)$$

The mandate is violated by the unconstrained allocation when

$$Q_t^{L,\text{unc}} \mu_B b_t^{L,B,\text{unc}} < m_t. \quad (53)$$

7.1 Why binding does not by itself imply a higher physical long-bond quantity

Because the mandate is stated in market-value rather than quantity terms, the inequality

$$Q_t^{L,\text{unc}} \mu_B b_t^{L,B,\text{unc}} < m_t \quad (54)$$

does not by itself imply

$$b_t^{L,B} > b_t^{L,B,\text{unc}}. \quad (55)$$

In the binding equilibrium, the relevant physical quantity is m_t/Q_t^L . Thus B 's physical long-bond quantity rises relative to the unconstrained allocation if and only if

$$\frac{m_t}{Q_t^L} > \mu_B b_t^{L,B,\text{unc}}, \quad (56)$$

or equivalently,

$$m_t > Q_t^L \mu_B b_t^{L,B,\text{unc}}. \quad (57)$$

This condition is stronger than unconstrained violation whenever $Q_t^L > Q_t^{L,\text{unc}}$.

7.2 Local quantity-response and share-response conditions

A differential sufficient condition in the binding region is

$$\frac{\partial b_t^{L,B}}{\partial m_t} > 0. \quad (58)$$

With fixed aggregate long-bond supply at date t ,

$$\mu_A b_t^{L,A} + \mu_B b_t^{L,B} = b_t^L, \quad (59)$$

so

$$\frac{\partial b_t^{L,A}}{\partial m_t} = -\frac{\mu_B}{\mu_A} \frac{\partial b_t^{L,B}}{\partial m_t} < 0. \quad (60)$$

To translate the fall in A 's long-bond level into a fall in its long-bond portfolio share, use

$$\alpha_t^A = \frac{b_t^{L,A}}{W_t^A}, \quad W_t^A = b_t^{S,A} + b_t^{L,A}. \quad (61)$$

Then

$$\frac{\partial \alpha_t^A}{\partial m_t} = \frac{1}{W_t^A} \frac{\partial b_t^{L,A}}{\partial m_t} - \frac{b_t^{L,A}}{(W_t^A)^2} \frac{\partial W_t^A}{\partial m_t}. \quad (62)$$

Thus

$$\frac{\partial \alpha_t^A}{\partial m_t} < 0 \quad (63)$$

if and only if

$$\frac{\partial b_t^{L,A}}{\partial m_t} < \alpha_t^A \frac{\partial W_t^A}{\partial m_t}. \quad (64)$$

Using $W_t^A = b_t^{S,A} + b_t^{L,A}$, this is equivalent to

$$(1 - \alpha_t^A) \frac{\partial b_t^{L,A}}{\partial m_t} < \alpha_t^A \frac{\partial b_t^{S,A}}{\partial m_t}. \quad (65)$$

A simple sufficient condition is

$$\frac{\partial b_t^{L,B}}{\partial m_t} > 0, \quad \frac{\partial b_t^{S,A}}{\partial m_t} \geq 0. \quad (66)$$

More generally, using long-bond market clearing, it is enough that

$$\frac{\partial b_t^{S,A}}{\partial m_t} > -\frac{1 - \alpha_t^A}{\alpha_t^A} \frac{\mu_B}{\mu_A} \frac{\partial b_t^{L,B}}{\partial m_t}. \quad (67)$$

Under these sufficient conditions, an increase in mandate intensity pushes agent A toward the short bond and lowers A 's long-bond portfolio share.

8 Fiscal rent and price-wedge stock revaluation

The leading-order fiscal rent is the wedge term that appears in the government present-value identity:

$$X_t \equiv \xi_t^{S,A} b_t^S + \xi_t^{L,A} b_t^L. \quad (68)$$

Substituting the wedge definitions,

$$X_t = \frac{\phi(\alpha_t^A - \theta_A)}{(c_t^A)^{-\sigma}} [\alpha_t^A b_t^S - (1 - \alpha_t^A) b_t^L]. \quad (69)$$

Using

$$\alpha_t^A = \frac{b_t^{L,A}}{W_t^A}, \quad 1 - \alpha_t^A = \frac{b_t^{S,A}}{W_t^A}, \quad (70)$$

we obtain

$$X_t = \frac{\phi(\alpha_t^A - \theta_A)}{(c_t^A)^{-\sigma} W_t^A} \left[b_t^{L,A} b_t^S - b_t^{S,A} b_t^L \right]. \quad (71)$$

Applying market clearing,

$$b_t^{L,A} b_t^S - b_t^{S,A} b_t^L = \mu_B \left(b_t^{L,A} b_t^{S,B} - b_t^{S,A} b_t^{L,B} \right), \quad (72)$$

so

$$X_t = \frac{\phi(\alpha_t^A - \theta_A) \mu_B}{(c_t^A)^{-\sigma} W_t^A} \left[b_t^{L,A} b_t^{S,B} - b_t^{S,A} b_t^{L,B} \right]. \quad (73)$$

This determinant representation is the clean algebraic object. It is exact within the leading-order wedge approximation and receives the second-order residual described below.

The price-wedge stock revaluation is distinct. Define the fundamental long-bond price as the unconstrained price,

$$Q_t^{L,\text{fund}} \equiv Q_t^{L,\text{unc}}, \quad (74)$$

and define

$$1 + \omega_t \equiv \frac{Q_t^L}{Q_t^{L,\text{fund}}}. \quad (75)$$

Then

$$\Psi_t \equiv (Q_t^L - Q_t^{L,\text{fund}}) b_t^L = \frac{\omega_t}{1 + \omega_t} Q_t^L b_t^L. \quad (76)$$

The objects X_t and Ψ_t are generically distinct. X_t is a flow-like present-value wedge constructed from agent A 's Euler-equation distortions and the cross-sectional portfolio determinant. Ψ_t is a stock revaluation of aggregate long-bond supply induced by the price wedge.

9 Government budget constraint and augmented FTPL identity

The government flow constraint is

$$\frac{b_{t-1}^S + (1 + \delta Q_t^L) b_{t-1}^L}{\pi_t} = s_t + Q_t^S b_t^S + Q_t^L b_t^L, \quad (77)$$

where $s_t \equiv \tau_t - g_t$ is the primary surplus. Define

$$v_t \equiv \frac{b_{t-1}^S + (1 + \delta Q_t^L) b_{t-1}^L}{\pi_t}, \quad \tilde{v}_t \equiv Q_t^S b_t^S + Q_t^L b_t^L. \quad (78)$$

Then

$$v_t = s_t + \tilde{v}_t. \quad (79)$$

At $t + 1$,

$$v_{t+1} = \frac{b_t^S + (1 + \delta Q_{t+1}^L) b_t^L}{\pi_{t+1}}. \quad (80)$$

Multiplying by agent A 's SDF and taking conditional expectations gives

$$\mathbb{E}_t[M_{t+1}^A v_{t+1}] = b_t^S \mathbb{E}_t \left[\frac{M_{t+1}^A}{\pi_{t+1}} \right] + b_t^L \mathbb{E}_t \left[\frac{M_{t+1}^A (1 + \delta Q_{t+1}^L)}{\pi_{t+1}} \right]. \quad (81)$$

Using agent A 's Euler equations with residuals,

$$\mathbb{E}_t[M_{t+1}^A v_{t+1}] = \tilde{v}_t - X_t - \mathcal{R}_t^A, \quad (82)$$

where

$$\mathcal{R}_t^A \equiv R_t^{S,A} b_t^S + R_t^{L,A} b_t^L. \quad (83)$$

Since $R_t^{S,A} = R_t^{L,A}$, we have

$$\mathcal{R}_t^A = -\frac{\phi}{2} \frac{(\varepsilon_t^A)^2}{(c_t^A)^{-\sigma}} (b_t^S + b_t^L) = O((\varepsilon_t^A)^2). \quad (84)$$

Combining with $v_t = s_t + \tilde{v}_t$,

$$v_t = s_t + X_t + \mathcal{R}_t^A + \mathbb{E}_t[M_{t+1}^A v_{t+1}]. \quad (85)$$

Forward iteration yields

$$v_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^j M_{t+k}^A \right) (s_{t+j} + X_{t+j} + \mathcal{R}_{t+j}^A), \quad (86)$$

provided the terminal term vanishes. At leading order,

$$v_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^j M_{t+k}^A \right) (s_{t+j} + X_{t+j}) + O(\varepsilon^2). \quad (87)$$

10 Local sign of the fiscal rent

Consider a neighborhood of the symmetric-supply unconstrained reference point where

$$\chi = \bar{\theta}, \quad (88)$$

so that, at the reference allocation,

$$\alpha^{A,\text{unc}} = \theta_A, \quad \alpha^{B,\text{unc}} = \theta_B. \quad (89)$$

At that point,

$$b^{L,A} b^{S,B} - b^{S,A} b^{L,B} = W^A W^B [\theta_A(1 - \theta_B) - (1 - \theta_A)\theta_B] \quad (90)$$

$$= W^A W^B (\theta_A - \theta_B) < 0. \quad (91)$$

By continuity, the determinant remains negative in a neighborhood.

Proposition 1 (Conditional local positivity). *Consider a binding-mandate equilibrium near the symmetric-supply reference point. Suppose the local quantity-response and share-response conditions hold:*

$$\frac{\partial b_t^{L,B}}{\partial m_t} > 0, \quad (92)$$

and

$$(1 - \alpha_t^A) \frac{\partial b_t^{L,A}}{\partial m_t} < \alpha_t^A \frac{\partial b_t^{S,A}}{\partial m_t}. \quad (93)$$

Then the mandate pushes agent A below its habitat,

$$\alpha_t^A < \theta_A, \quad (94)$$

and the leading-order fiscal rent satisfies

$$X_t > 0. \quad (95)$$

Including the exact second-order habitat residual, the present-value contribution is $X_t + \mathcal{R}_t^A$, where

$$\mathcal{R}_t^A = -\frac{\phi}{2} \frac{(\alpha_t^A - \theta_A)^2}{(c_t^A)^{-\sigma}} (b_t^S + b_t^L) < 0. \quad (96)$$

Thus the leading-order rent is positive locally, while the residual is negative and second order.

11 Transversality

The present-value identity requires

$$\lim_{T \rightarrow \infty} \mathbb{E}_t \left[\left(\prod_{k=1}^{T+1} M_{t+k}^A \right) v_{t+T+1} \right] = 0. \quad (97)$$

Agent A 's household transversality gives

$$\lim_{T \rightarrow \infty} \mathbb{E}_t \left[\left(\prod_{k=1}^{T+1} M_{t+k}^A \right) \left(Q_{t+T}^S b_{t+T}^{S,A} + Q_{t+T}^L b_{t+T}^{L,A} \right) \right] = 0. \quad (98)$$

For agent B , the bounded-mandate condition controls the market value of mandated long-bond holdings, but does not by itself control short-bond positions. This is why the admissibility condition on bounded gross positions or natural debt limits is required. Under that condition and standard regularity on the endowment process, for example that y_t grows slowly enough under agent A 's SDF, the discounted value of total government debt satisfies the required transversality condition.

12 Dynamic linearization convention

For impulse-response analysis it is convenient to linearize around a deterministic steady state with symmetric per-capita endowments and symmetric initial portfolios. In that case one may impose

$$W^A = W^B, \quad (99)$$

which implies

$$\omega_i = \mu_i. \quad (100)$$

Under the additional symmetry in endowments and initial wealth, one also has

$$c^A = c^B. \quad (101)$$

The equality $W^A = W^B$ alone does not imply $c^A = c^B$; the latter requires the additional symmetry just stated. With these conventions, the common-consumption closed-form portfolio-share formulas apply directly to the linearized model.

13 What remains intentionally not claimed

1. **Global monotonicity is not established here.** A global claim that X_t is strictly increasing in mandate intensity requires additional restrictions on the price response $\partial Q_t^L / \partial m_t$, the physical quantity response $\partial b_t^{L,B} / \partial m_t$, and the response of A 's total portfolio wealth.
2. **The positivity of X_t is local and conditional.** It relies on the determinant remaining negative near the symmetric-supply reference point and on the sufficient conditions ensuring that A 's long-bond share falls below habitat.
3. **X_t and Ψ_t are generically distinct.** They answer different questions: X_t is the Euler-wedge object entering the present-value identity, while Ψ_t is a stock revaluation. A formal measure-zero equivalence statement would require a separate proof.
4. **Bounded gross positions are an admissibility assumption.** They are not implied by the budget constraint alone and should be stated explicitly if used for transversality.

14 Summary of corrected main results

The leading-order fiscal rent is

$$X_t = \frac{\phi(\alpha_t^A - \theta_A)\mu_B}{(c_t^A)^{-\sigma}W_t^A} \left[b_t^{L,A}b_t^{S,B} - b_t^{S,A}b_t^{L,B} \right]. \quad (102)$$

The corrected present-value identity is

$$v_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^j M_{t+k}^A \right) (s_{t+j} + X_{t+j} + \mathcal{R}_{t+j}^A), \quad (103)$$

with

$$\mathcal{R}_t^A = -\frac{\phi(\alpha_t^A - \theta_A)^2}{2(c_t^A)^{-\sigma}} (b_t^S + b_t^L) = O((\alpha_t^A - \theta_A)^2). \quad (104)$$

At leading order,

$$v_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^j M_{t+k}^A \right) (s_{t+j} + X_{t+j}) + O(\varepsilon^2). \quad (105)$$

The common-consumption portfolio-share deviations are

$$\varepsilon^A = \frac{(\chi - \bar{\theta})(1 - \theta_B)}{\omega_A(1 - \theta_B) + \omega_B(1 - \theta_A)}, \quad \varepsilon^B = \frac{(\chi - \bar{\theta})(1 - \theta_A)}{\omega_A(1 - \theta_B) + \omega_B(1 - \theta_A)}. \quad (106)$$

The consumption-corrected formulas are

$$\varepsilon^A = \frac{(\chi - \bar{\theta})(1 - \theta_B)(c^B)^\sigma}{\omega_A(1 - \theta_B)(c^B)^\sigma + \omega_B(1 - \theta_A)(c^A)^\sigma}, \quad (107)$$

$$\varepsilon^B = \frac{(\chi - \bar{\theta})(1 - \theta_A)(c^A)^\sigma}{\omega_A(1 - \theta_B)(c^B)^\sigma + \omega_B(1 - \theta_A)(c^A)^\sigma}. \quad (108)$$

Finally, the binding market-value mandate raises B 's physical long-bond holding only under the strengthened condition

$$\frac{m_t}{Q_t^L} > \mu_B b_t^{L,B,\text{unc}}, \quad (109)$$

or locally under

$$\frac{\partial b_t^{L,B}}{\partial m_t} > 0. \quad (110)$$

It lowers A 's long-bond portfolio share if, in addition,

$$(1 - \alpha_t^A) \frac{\partial b_t^{L,A}}{\partial m_t} < \alpha_t^A \frac{\partial b_t^{S,A}}{\partial m_t}, \quad (111)$$

with the simple sufficient condition

$$\frac{\partial b_t^{S,A}}{\partial m_t} \geq 0. \quad (112)$$